

LAPORAN TUGAS PEMROGRAMAN BERBASIS WEB

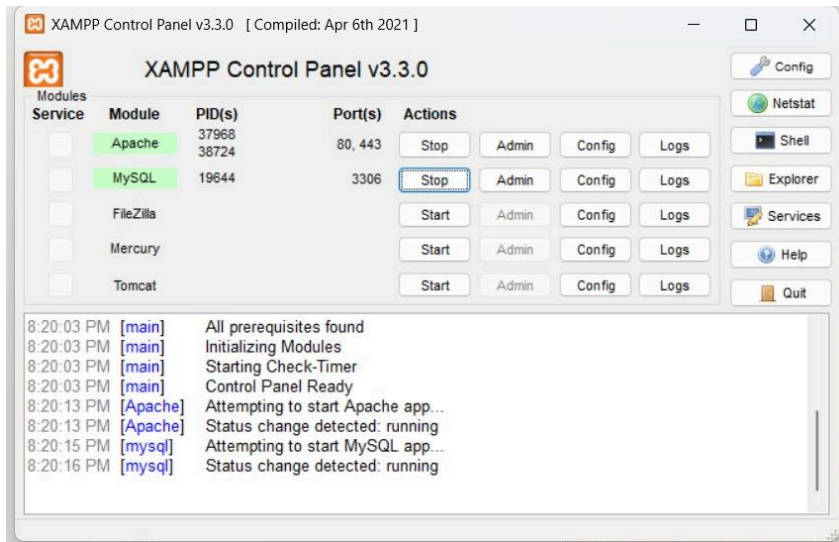
AKTIVITAS MANDIRI 3 DAN 4

NAMA : ADINE VIVIA

NIM : 242410101083

KELAS : PWEB A

AKTIVITAS MANDIRI 3



1. Membuat Controller

Buatlah sebuah Controller baru dengan nama DashboardController menggunakan perintah artisan

```
.env | DashboardController.php | web.php | app.blade.php | navbar.blade.php | ter
app > Http > Controllers > DashboardController.php
3 namespace App\Http\Controllers;
4
5 use Illuminate\Http\Request;
6
7 class DashboardController extends Controller
8 {
9     public function index()
10     {
11         return view('dashboard', ['data' => ['Jumlah Mahasiswa: 100', 'Mata Kuliah: 12']]);
12     }
13 }
14
```

2. Membuat Route

Daftarkan Route baru yang mengarahkan URL /dashboard ke method index di dalam DashboardController

```
.env DashboardController.php X web.php app.blade.php navbar.blade.php ter
app > Http > Controllers > DashboardController.php
3 namespace App\Http\Controllers;
4
5 use Illuminate\Http\Request;
6
7 class DashboardController extends Controller
8 {
9     public function index()
10     {
11         return view('dashboard', ['data' => ['Jumlah Mahasiswa: 100', 'Mata Kuliah: 12']]);
12     }
13 }
14
```

```
2026_05_05_130000_create_categories_table.php.php web.php X
routes > web.php
1 <?php
2
3 use App\Http\Controllers\DashboardController;
4 use Illuminate\Support\Facades\Route;
5
6 Route::get('/', function () {
7     return view('welcome');
8 });
9
10 Route::get('/dashboard', [DashboardController::class,
11 Route::get('/hitung/{a}/{b}', function ($a, $b) {
12     return $a + $b;
13 });
14 Route::view('/tentang', 'tentang');
```

3. Mengirim Data ke View

Buatlah method index di dalam DashboardController yang mengirimkan data (bisa berupa array atau object) ke sebuah file view

```
.env DashboardController.php web.php app.blade.php navbar.blade.php X tentang.blade.php
resources > views > partials > navbar.blade.php > nav
1 <nav><b>Navbar Vi's Yum</b> | <a href="/dashboard">Dashboard</a> | <a href="/tentang">Tentang</a></nav>
```

4. Membuat View dengan Blade

Buatlah file view baru bernama dashboard.blade.php di dalam folder resources/views/

```
resources > views > dashboard.blade.php > ...
1  @extends('layouts.app')
2  @section('content')
3      <h1>Dashboard Vi's Yum</h1>
4      @foreach($data as $item)
5          <li>{{ $item }}</li>
6      @empty
7          <p>Data Kosong</p>
8      @endforeach
9  @endsection
```

5. Implementasi Blade Directives

Gunakan directive Blade `@if`, `@foreach`, dan `@empty` untuk menampilkan data yang dikirim dari Controller ke dalam View dashboard

```
resources > views > dashboard.blade.php > ...
1  @extends('layouts.app')
2  @section('content')
3      <h1>Dashboard Vi's Yum</h1>
4      @foreach($data as $item)
5          <li>{{ $item }}</li>
6      @empty
7          <p>Data Kosong</p>
8      @endforeach
9  @endsection
```

6. Integrasi Template (Layouting)

Implementasikan *template inheritance* dengan menggunakan `@extends` dan `@section` agar halaman dashboard mengikuti layout utama aplikasi

```
resources > views > dashboard.blade.php > ...
1 @extends('layouts.app')
2 @section('content')
3     <h1>Dashboard Vi's Yum</h1>
4     @forelse($data as $item)
5         <li>{{ $item }}</li>
6     @empty
7         <p>Data Kosong</p>
8     @endforelse
9 @endsection
```

Dashboard Vi's Yum

- Jumlah Mahasiswa: 100
- Mata Kuliah: 12

© 2024 Vi's Yum

AKTIVITAS MANDIRI 4

1. Membuat Migration

Buatlah file migration baru untuk membuat tabel php artisan make:model Mahasiswa -m

```
PS C:\xampp\htdocs\visyum-app> php artisan make:model Menu -m

INFO Model [C:\xampp\htdocs\visyum-app\app\Models\Menu.php] created successfully.

INFO Migration [C:\xampp\htdocs\visyum-app\database\Migrations\2026_05_05_134548_create_menus_table.php] created successfully.

PS C:\xampp\htdocs\visyum-app>
```

```
database > migrations > 2026_05_05_134548_create_menus_table.php
4 use Illuminate\Database\Schema\Blueprint;
5 use Illuminate\Support\Facades\Schema;
6
7 return new class extends Migration
8 {
9     /**
10      * Run the migrations.
11      */
12     public function up(): void
13     {
14         Schema::create('menus', function (Blueprint $table) {
15             $table->id();
16             $table->string('kode_menu')->unique(); // Pengganti NIM
17             $table->string('nama_menu'); // Pengganti Nama
18             $table->enum('kategori', ['Makanan', 'Minuman', 'Snack']); // Pengganti Jurusan (Enum)
19             $table->decimal('harga', 10, 2); // Pengganti IPK (Decimal)
20             $table->integer('stok'); // Pengganti Semester (Integer)
21             $table->boolean('tersedia'); // Pengganti Aktif (Boolean)
22             $table->string('foto')->nullable(); // Tetap Foto
23             $table->timestamps();
24         });
25     }
26
27     /**
28      * Reverse the migrations.
29      */
30     public function down(): void
31     {
32         Schema::dropIfExists('menus');
33     }
34 };
35
```

2. Menjalankan Migration

Jalankan perintah migration untuk membentuk tabel tersebut di dalam database MySQL (phpMyAdmin)

```
PS C:\xampp\htdocs\visyum-app> php artisan migrate

INFO Running migrations.

2026_05_05_134548_create_menus_table ..... 303.86ms DONE

PS C:\xampp\htdocs\visyum-app>
```

3. Membuat Model

Buatlah sebuah Model bernama Menu yang merepresentasikan tabel menus dan atur properti \$fillable di dalamnya

```

app > Models > Menu.php
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 class Menu extends Model
8 {
9     protected $fillable = [
10         'kode_menu',
11         'nama_menu',
12         'kategori',
13         'harga',
14         'stok',
15         'tersedia',
16         'foto'
17     ];
18
19     protected $casts = [
20         'tersedia' => 'boolean',
21         'harga' => 'decimal:2',
22     ];
23
24     public function scopeTersedia($query)
25     {
26         return $query->where('tersedia', true);
27     }
28 }

```

4. Implementasi Seeder

Buatlah file Seeder baru untuk mengisi tabel menus dengan data awal (dummy data) minimal sebanyak 5-6 data menu

```

PS C:\xampp\htdocs\visyum-app> php artisan make:seeder MenuSeeder

INFO Seeder [C:\xampp\htdocs\visyum-app\database\seeders\MenuSeeder.php] created successfully.

PS C:\xampp\htdocs\visyum-app>

```

<pre> database > seeders > MenuSeeder.php 1 <?php 2 3 namespace Database\Seeders; 4 5 use Illuminate\Database\Seeder; 6 use App\Models\Menu; 7 8 class MenuSeeder extends Seeder 9 { 10 public function run(): void 11 { 12 \App\Models\Menu::create([13 'kode_menu' => 'VIS001', 14 'nama_menu' => 'Donat Gula Klasik', 15 'kategori' => 'Snack', 16 'harga' => 3000, 17 'stok' => 50, 18 'tersedia' => true 19]); 20 21 \App\Models\Menu::create([22 'kode_menu' => 'VIS002', 23 'nama_menu' => 'Lempeng Ayam Bakar', 24 'kategori' => 'Snack', 25 'harga' => 4000, 26 'stok' => 40, 27 'tersedia' => true 28]); </pre>	<pre> database > seeders > MenuSeeder.php 9 { 10 { 11 12 \App\Models\Menu::create([13 'kode_menu' => 'VIS003', 14 'nama_menu' => 'Pastel Sayur Renyah', 15 'kategori' => 'Snack', 16 'harga' => 3500, 17 'stok' => 30, 18 'tersedia' => true 19]); 20 21 \App\Models\Menu::create([22 'kode_menu' => 'VIS004', 23 'nama_menu' => 'Kue Lapis Kanji', 24 'kategori' => 'Snack', 25 'harga' => 2500, 26 'stok' => 45, 27 'tersedia' => true 28]); 29 30 \App\Models\Menu::create([31 'kode_menu' => 'VIS005', 32 'nama_menu' => 'Risoles Mayo', 33 'kategori' => 'Snack', 34 'harga' => 5000, 35 'stok' => 20, 36 'tersedia' => true 37]); 38 39 } 40 } </pre>
--	---

```

9      {
11         {
57             \App\Models\Menu::create([
58                 'kode_menu' => 'VIS006',
59                 'nama_menu' => 'Klepon Gula Merah',
60                 'kategori' => 'Snack',
61                 'harga' => 5000,
62                 'stok' => 25,
63                 'tersedia' => true
64             ]);
65         }
66     }

```

				id	category_id	kode_menu	nama_menu	harga	stok	tersedia	created_at	updated_at
☐	Edit	Copy	Delete	1	1	VIS001	Donat Gula Klasik	3000.00	50	1	2026-05-05 14:20:13	2026-05-05 14:20:13
☐	Edit	Copy	Delete	2	1	VIS002	Lemper Ayam Bakar	4000.00	40	1	2026-05-05 14:20:13	2026-05-05 14:20:13
☐	Edit	Copy	Delete	3	1	VIS006	Klepon Gula Merah	5000.00	25	1	2026-05-05 14:20:13	2026-05-05 14:20:13
☐	Edit	Copy	Delete	4	2	VIS003	Pastel Sayur Renyah	3500.00	30	1	2026-05-05 14:20:13	2026-05-05 14:20:13
☐	Edit	Copy	Delete	5	2	VIS004	Kue Lapis Kanji	2500.00	45	1	2026-05-05 14:20:13	2026-05-05 14:20:13
☐	Edit	Copy	Delete	6	2	VIS005	Risoles Mayo	5000.00	20	1	2026-05-05 14:20:13	2026-05-05 14:20:13

5. Eksekusi SeedeR

Jalankan perintah Seeder melalui terminal untuk memasukkan data ke dalam database tanpa melalui input manual di phpMyAdmin

```

PS C:\xampp\htdocs\visyum-app> php artisan migrate

INFO Running migrations.

2026_05_05_134548_create_menus_table ..... 303.86ms DONE

PS C:\xampp\htdocs\visyum-app>

```



```

PS C:\xampp\htdocs\visyum-app> php artisan migrate:fresh --seed

Dropping all tables ..... 191.49ms DONE

INFO Preparing database.

Creating migration table ..... 19.38ms DONE

INFO Running migrations.

0001_01_01_000000_create_users_table ..... 111.29ms DONE
0001_01_01_000001_create_cache_table ..... 73.25ms DONE
0001_01_01_000002_create_jobs_table ..... 119.06ms DONE
2026_05_05_134548_create_menus_table ..... 25.33ms DONE

INFO Seeding database.

Database\Seeders\MenuSeeder ..... RUNNING
Database\Seeders\MenuSeeder ..... 162 ms DONE

PS C:\xampp\htdocs\visyum-app>

```

6. Uji Coba Query (Eloquent)

Gunakan fitur Laravel Tinker untuk menjalankan query Eloquent seperti `all()`, `find()`, atau `where()` guna memastikan data berhasil dipanggil dari database

```

PS C:\xampp\htdocs\visyum-app> php artisan tinker
Psy Shell v0.12.22 (PHP 8.2.12 - cli) by Justin Hileman
New PHP manual is available (latest: 3.0.5). Update with `doc --update-manual`

>

```

- `App\Models\Menu::all();`

```

> App\Models\Menu::all();

Illuminate\Database\Eloquent\Collection {#852}
  all: [
    App\Models\Menu {#8071}
      id: 1,
      kode_menu: "VIS001",
      nama_menu: "Donat Gula Klasik",
      kategori: "Snack",
      harga: "3000.00",
      stok: 50,
      tersedia: 1,
      foto: null,
      created_at: "2026-05-05 13:58:01",
      updated_at: "2026-05-05 13:58:01",
    ],
    App\Models\Menu {#8072}
      id: 2,
      kode_menu: "VIS002",
      nama_menu: "Lemper Ayam Bakar",
      kategori: "Snack",
      harga: "4000.00",
      stok: 40,
      tersedia: 1,
      foto: null,
      created_at: "2026-05-05 13:58:01",
      updated_at: "2026-05-05 13:58:01",
  ]

```

```

    App\Models\Menu {#8073}
      id: 3,
      kode_menu: "VIS003",
      nama_menu: "Pastel Sayur Renyah",
      kategori: "Snack",
      harga: "3500.00",
      stok: 30,
      tersedia: 1,
      foto: null,
      created_at: "2026-05-05 13:58:01",
      updated_at: "2026-05-05 13:58:01",
    ],
    App\Models\Menu {#8074}
      id: 4,
      kode_menu: "VIS004",
      nama_menu: "Kue Lapis Kanji",
      kategori: "Snack",
      harga: "2500.00",
      stok: 45,
      tersedia: 1,
      foto: null,
      created_at: "2026-05-05 13:58:01",
      updated_at: "2026-05-05 13:58:01",
  ]

```



```

    },
    App\Models\Menu {#8075
      id: 5,
      kode_menu: "VIS005",
      nama_menu: "Risoles Mayo",
      kategori: "Snack",
      harga: "5000.00",
      stok: 20,
      tersedia: 1,
      foto: null,
      created_at: "2026-05-05 13:58:01",
      updated_at: "2026-05-05 13:58:01",
    },
    App\Models\Menu {#8076
      id: 6,
      kode_menu: "VIS006",
      nama_menu: "Klepon Gula Merah",
      kategori: "Snack",
      harga: "5000.00",
      stok: 25,
      tersedia: 1,
      foto: null,
      created_at: "2026-05-05 13:58:01",
      updated_at: "2026-05-05 13:58:01",
    },
  ],
],

```

- App\Models\Menu::where('kategori', 'Snack')->get();

```

> 'Snack')->get();

= Illuminate\Database\Eloquent\Collection {#7374
  all: [
    App\Models\Menu {#8109
      id: 1,
      kode_menu: "VIS001",
      nama_menu: "Donat Gula Klasik",
      kategori: "Snack",
      harga: "3000.00",
      stok: 50,
      tersedia: 1,
      foto: null,
      created_at: "2026-05-05 13:58:01",
      updated_at: "2026-05-05 13:58:01",
    },
    App\Models\Menu {#8110
      id: 2,
      kode_menu: "VIS002",
      nama_menu: "Lemper Ayam Bakar",
      kategori: "Snack",
      harga: "4000.00",
      stok: 40,
      tersedia: 1,
      foto: null,
      created_at: "2026-05-05 13:58:01",
      updated_at: "2026-05-05 13:58:01",
    },
    App\Models\Menu {#8111
      id: 3,
      kode_menu: "VIS003",
      nama_menu: "Pastel Sayur Renyah",
      kategori: "Snack",
      harga: "3500.00",
      stok: 30,
      tersedia: 1,
    },
  ],
}

```

- App\Models\Menu::avg('harga');

```
> App\Models\Menu::avg('harga');  
  
= "3833.333333"  
  
> █
```

7. Membuat Relasi Antar Tabel

Buatlah relasi database (One-to-Many atau Many-to-Many). Contoh: Relasi antara tabel categories dan tabel menus, serta pastikan relasi tersebut berjalan dengan benar di level Model dan Database (Foreign Key)

```
PS C:\xampp\htdocs\visyum-app> php artisan make:model Category -m  
  
INFO Model [C:\xampp\htdocs\visyum-app\app\Models\Category.php] created successfully.  
  
INFO Migration [C:\xampp\htdocs\visyum-app\database\Migrations\2026_05_05_140842_create_categories_table.php] created successfully.
```

```
database > migrations > 2026_05_05_140842_create_categories_table.php  
1 <?php  
2  
3 use Illuminate\Database\Migrations\Migration;  
4 use Illuminate\Database\Schema\Blueprint;  
5 use Illuminate\Support\Facades\Schema;  
6  
7 return new class extends Migration  
8 {  
9     public function up(): void  
10     {  
11         Schema::create('categories', function (Blueprint $table) {  
12             $table->id();  
13             $table->string('nama_kategori');  
14             $table->timestamps();  
15         });  
16     }  
17  
18     public function down(): void  
19     {  
20         Schema::dropIfExists('categories');  
21     }  
22 };
```

```
2026_05_05_134548_create_menus_table.php Category.php X  
app > Models > Category.php  
1 <?php  
2  
3 namespace App\Models;  
4  
5 use Illuminate\Database\Eloquent\Model;  
6 use Illuminate\Database\Eloquent\Factories\HasFactory;  
7  
8 class Category extends Model  
9 {  
10     use HasFactory;  
11  
12     protected $fillable = ['nama_kategori'];  
13  
14     public function menus()  
15     {  
16         return $this->hasMany(Menu::class);  
17     }  
18 }
```